UNIT 1
COMPUTER SYSTEMS

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AGENDA

MACHINE LEVEL REPRESENTATION OF DATA

- "Machine level representation of data" refers to how information is encoded and stored in a computer's memory, which is understood by the machine's hardware. At this level, data is represented using binary digits (bits), which are either 0s or 1s.

- **Binary Representation:**
  - At the machine level, all data is represented using binary digits. A binary digit, or a bit, can have one of two states: 0 or 1. These bits are the fundamental units of information in a computer. For example:
  - The decimal number 5 is represented in binary as 101.
  - The ASCII character 'A' is represented as 01000001 in binary.
  - A color in an image might be represented as a combination of binary values for red, green, and blue components.

DATA TYPES

- Different types of data are represented in different ways at the machine level. Common data types include integers, floating-point numbers, characters, and Boolean values.
- **Integers:** Integers are whole number (no fractions or decimals), e.g., -3, 0, 7. Int has size of 4 bytes (32bit).
- **Floating-Point Numbers:** A floating point number, is a positive or negative whole number with a decimal point. For example, 15.5, 0.45, and -203.345.
- **Characters:** Characters are represented using character encoding schemes such as ASCII or Unicode, where each character is assigned a unique binary code. For example, the ASCII code for the letter 'A' is 65 (01000001 in binary).
- **Boolean Values:** Boolean values, representing true or false, can be represented using a single bit, where 0 typically represents false and 1 represents true.

NUMBER SYSTEM

- A number is a way of writing and showing numbers.
- It uses specific group of digits to represent particular number system.
- In this unit scope, we will study below number systems:
  - Decimal Number System (base-10)
  - Binary Number System (base-2)
  - Octal Number System (base-8)
  - Hexadecimal Number System (base-16)

NUMBER SYSTEM: DECIMAL NUMBER SYSTEM

- Also known as denary number system which is composed of 10 symbols: 0,1,2,3,4,5,6,7,8,9
- Also called base-10 because it has 10 digits.
- It is a positional-value system, in which the value of a digit depends on its position in the number
- For example, consider the decimal number 964.
  - 9 represents 9 hundreds
  - 6 represents 6 tens
  - 4 represents 4 units
  - The 9 carries the most weight. It is called the Most Significant Digit (MSD). The 4 carries the least weight and is called the Least Significant Digit (LSD).

NUMBER SYSTEM: BINARY NUMBER SYSTEM

- In binary number system there are only two symbols or digits, 0 and 1.
- The base of binary number system is 2. The binary system is also a positional-value system like decimal number system, wherein each bit has its own value or weight expressed as power of 2.
- For example:
  - $1011.101_2 = 1X2^3 + 0X2^2 + 1X2^1 + 1X2^0 + 1X2^{-1} + 0X2^{-2} + 1X2^{-3}$
  - $= 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$
  - $= 11.625_{10}$

NUMBER SYSTEM: OCTAL NUMBER SYSTEM

- The Octal number system has a base of 8, meaning it has eight possible digits; 0, 1, 2, 3, 4, 5, 6, and 7.
- Thus, each digit of an octal number system can have any value from 0 to 7.
- The digit positions in an octal number system have weight with powers of 8.
- An octal number can be easily converted to its decimal equivalent by multiplying each digit by its positional weight.
- For example,
  - $(672)_8 = 6x8^2 + 7x8^1 + 2x8^0 = 384 + 56 + 2 = (442)_{10}$

NUMBER SYSTEM: HEXADECIMAL NUMBER SYSTEM

- The hexadecimal number system uses base 16.
- It uses the digits 0 - 9 and letters A, B, C, D, E and F as the 16-digit symbols.
- The table shows the relationship among hexadecimal and decimal numbers.
- A hexadecimal number can be converted to its decimal equivalent using below way
  - $3AF_{16} = 3x16^2 + 10x16^1 + 15x16^0 = 768 + 160 + 15$
  - $= 943_{10}$

HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
DEC	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

CONVERSION FROM BINARY TO DECIMAL

- The process involves multiplying each bit by its positional weight (powers of 2) and adding the results.
- **Example 1: Whole Binary Number** = Convert $(1101)_2$ to decimal.
  - $1\times2^3 = 1\times8 = 8$
  - $1\times2^2 = 1\times4 = 4$
  - $0\times2^1 = 0\times2 = 0$
  - $1\times2^0 = 1\times1 = 1$
  - **Sum:** $8 + 4 + 0 + 1 = (\mathbf{13})_{\mathbf{10}}$

CONVERSION FROM BINARY TO DECIMAL CONT.

- **Example 2: Binary with a Fraction:** Convert $(1011.101)_2$ to decimal.
  - $1\times2^3 = 8$
  - $0\times2^2 = 0$
  - $1\times2^1 = 2$
  - $1\times2^0 = 1$
  - $1\times2^{-1} = 0.5$
  - $0\times2^{-2} = 0$
  - $1\times2^{-3} = 0.125$
  - **Sum:** $8 + 0 + 2 + 1 + 0.5 + 0 + 0.125 = (\mathbf{11.625})_{\mathbf{10}}$

CONVERSION FROM BINARY TO DECIMAL CONT.

- **Practice Problems:** Try converting these binary numbers to decimal.
- **Set A: Whole Numbers**
  - $(101)_2$
  - $(1110)_2$
  - $(10001)_2$
  - $(110110)_2$
- **Set B: Fractional Numbers**
  - $(11.1)_2$
  - $(101.01)_2$
  - $(1101.11)_2$

CONVERSION FROM DECIMAL TO BINARY.

- To convert a whole decimal number to binary, divide the number by 2 repeatedly, keeping track of the remainders. The binary number is read from the bottom remainder to the top remainder.

- **Example 1: Convert $(13)_{10}$ to binary**
  - $13 \div 2 = 6$, Remainder **1** (Least Significant Bit / LSD)
  - $6 \div 2 = 3$, Remainder **0**
  - $3 \div 2 = 1$, Remainder **1**
  - $1 \div 2 = 0$, Remainder **1** (Most Significant Bit / MSB)
  - *Read from bottom to top:* $(1101)_2$

CONVERSION FROM DECIMAL TO BINARY.

- **Example 2: Convert $(123.54)_{10}$ to binary**

• **Step 1: Convert the Whole Number (123)** • We use **repeated division by 2.** • $123 \div 2 = 61$, Remainder **1** • $61 \div 2 = 30$, Remainder **1** • $30 \div 2 = 15$, Remainder **0** • $15 \div 2 = 7$, Remainder **1** • $7 \div 2 = 3$, Remainder **1** • $3 \div 2 = 1$, Remainder **1** • $1 \div 2 = 0$, Remainder **1** • Read from **bottom to top:** $(123)_{10} = (1111011)_2$	**Step 2: Convert the Fraction (0.54)** We use **repeated multiplication by 2.** We will multiply, pull down the whole number (0 or 1), and carry the remaining decimal over to the next step. $0.54 \times 2 = \mathbf{1.08}$ →Write down **1.** (Carry the .08) $0.08 \times 2 = \mathbf{0.16}$ →Write down **0.** (Carry the .16) $0.16 \times 2 = \mathbf{0.32}$ →Write down **0.** (Carry the .32) $0.32 \times 2 = \mathbf{0.64}$ →Write down **0.** (Carry the .64) $0.64 \times 2 = \mathbf{1.28}$ →Write down **1.** (Carry the .28) $0.28 \times 2 = \mathbf{0.56}$ →Write down **0.** (Carry the .56) Read from **top to bottom:** $(0.54)_{10} \approx (0.100010\ldots)_2$(We'll stop here for precision).

CONVERSION FROM DECIMAL TO BINARY CONT.

- **Practice Problems:** Try converting these decimal numbers to binary.

Whole Numbers	Fractional Numbers	Mixed Numbers
• $(7)_{10}$ • $(15)_{10}$ • $(21)_{10}$ • $(50)_{10}$ • $(64)_{10}$	• *Use the repeated multiplication by 2 method. Stop when the fraction becomes 0, or after 4 binary places if it repeats.* • $(0.5)_{10}$ • $(0.875)_{10}$ • $(0.2)_{10}$ • *(Hint: This will repeat. Calculate to 4 binary places).*	• $(6.5)_{10}$ • $(15.25)_{10}$ • $(2.75)_{10}$

CONVERSION FROM OCTAL TO BINARY

- Converting **Octal to Binary** is actually the easiest conversion in computer science! Because 8is a power of $2(2^3 = 8)$, you don't need to do any complex math.
- You just convert each octal digit into its **3-bit binary equivalent** (sometimes called a "triad").
- **The "Magic" Cheat Sheet**

0	1	2	3	4	5	6	7
000	001	010	011	100	101	110	111

CONVERSION FROM OCTAL TO BINARY CONT.

Example 1: Simple Conversion	Example 2: Multi-digit	Example 3: With a fraction (Octal Point)
• Convert $(7)_8$ to binary. • Look at the chart: 7 is 111. • Answer: $(111)_2$	• Convert $(25)_8$ to binary. • Convert the 2: 2 = 010 • Convert the 5: 5 = 101 • *Put them together:* $(010101)_2$ *(or just drop the leading zero to make it* $(10101)_2$*).*	• Convert $(64.3)_8$ to binary. • Convert the 6: 6 = 110 • Convert the 4: 4 = 100 • Convert the 3: 3 = 011 • Combine with a binary point: $(110100.011)_2$

CONVERSION FROM OCTAL TO BINARY CONT.

- **Practice Problems:** Try converting these octal numbers to binary.

Whole Numbers	Fractional Numbers
$(17)_8$ $(5)_8$ $(44)_8$ $(72)_8$ $(305)_8$	$(5.2)_8$ $(12.4)_8$

CONVERSION FROM BINARY TO OCTAL

- Converting **Binary to Octal** is just the reverse of what we just did. You group the binary digits into sets of **three (3)**, starting from the decimal point and moving outwards, then convert each group into a single octal digit.
- **The Rules:**
  - **Group by 3:** Starting at the binary point (or the right side if there's no point), group the bits into threes.
  - **Pad with Zeros:** If you don't have enough digits at the far left (for whole numbers) or far right (for fractions), add zeros to make a full group of three.
  - **Convert:** Use the cheat sheet to translate each 3-bit group into its octal value.

000	001	010	011	100	101	110	111
0	1	2	3	4	5	6	7

CONVERSION FROM BINARY TO OCTAL CONT.

Example 1: Simple Conversion	Example 2: Needs Padding	Example 3: With a Fraction
• ***Convert*** $(\mathbf{101101})_2$***to octal.*** • Group by 3 from the right: **101** | **101** • Convert: 101 = **5**, 101 = **5** • Answer: $(55)_8$	• ***Convert*** $(\mathbf{1101})_2$***to octal.*** • Group from right: **1** | **101** • The left group only has one digit. Pad it with zeros: **001** | **101** • Convert: 001 = **1**, 101 = **5** • Answer: $(15)_8$	• ***Convert*** $(\mathbf{1101.10})_2$***to octal*** • Whole part (group from left of the point): **001** | **101** (Added two zeros to the left) • Fraction part (group from right of the point): **10** →pad with zero →**100** • Convert: 001 = **1**, 101 = **5** . 100 = **4** • Answer: $(15.4)_8$

CONVERSION FROM BINARY TO OCTAL CONT.

- **Practice Problems:** Try converting these octal numbers to binary. *(Hint: Watch out for padding!)*

Whole Numbers	Fractional Numbers
$(111)_2$ $(110010)_2$ $(10101)_2$ $(100000)_2$ $(1111000011)_2$	$(101.01)_2$ $(1110.111)_2$

CONVERSION FROM HEXADECIMAL TO BINARY

- Just like Octal, converting **Hexadecimal to Binary** is incredibly easy because of a shortcut. Since $16$is a power of $2(2^4 = 16)$, you simply convert each hexadecimal digit into its **4-bit binary equivalent** (sometimes called a "nibble").
- Cheat Sheet:

Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bin	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

CONVERSION FROM HEXADECIMAL TO BINARY

Example 1: Whole Number	Example 2: Using Letters	Example 3: With a Fraction (Hex Point)
• Convert $(5F)_{16}$ to binary. • Convert the **5**: 5 = **0101** • Convert the **F**: F (15) = **1111** • Put them together: $(01011111)_2$ (or drop the leading zero to make it $(1011111)_2$).	• Convert $(A2)_{16}$ to binary. • Convert the **A**: A (10) = **1010** • Convert the **2**: 2 = **0010** • Put them together: $(10100010)_2$	• Convert $(1C.8)_{16}$ to binary. • Convert the **1**: 1 = **0001** • Convert the **C**: C (12) = **1100** • Convert the **8**: 8 = **1000** • Combine with a binary point: $(00011100.1000)_2$ (or $(11100.1)_2$ if you drop the outer zeros).

CONVERSION FROM HEXADECIMAL TO BINARY

- **Practice Problem:** Convert these hexadecimal numbers to binary. *(Remember: Always write exactly 4 bits for each hex digit).*

Whole Numbers	Fractional Numbers
$(3)_{16}$ $(B)_{16}$ $(45)_{16}$ $(9A)_{16}$ $(1F0)_{16}$	$(D.4)_{16}$ $(A.1B)_{16}$

CONVERSION FROM BINARY TO HEX

- Converting **Binary to Hexadecimal** is the exact reverse of what we just did. You group the binary digits into sets of **four (4)**, starting from the binary point and moving outwards, and then translate each group using the cheat sheet.
- **The Rules:**
- **Group by 4:** Starting at the binary point, group the bits into sets of four.
- **Pad with Zeros:** If the outer groups (far left or far right) don't have 4 digits, pad them with zeros.
- **Convert:** Translate each 4-bit group into its Hex equivalent.
- *(Using the horizontal table from earlier for reference!)*

Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bin	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

CONVERSION FROM BINARY TO HEX

Example 1: Whole Number	Example 2: Another Whole Number	Example 3: With a Fraction
• Convert $(1011010)_2$ to Hex. • Group by 4 from the right: **101** | **1010** • The left group only has 3 digits. Pad it with a zero: **0101** | **1010** • Convert: 0101 = **5**, 1010 = **A** • Answer: $(5A)_{16}$	• Convert $(11010011)_2$ to Hex. • Group by 4 from the right: **1101** | **0011** • Convert: 1101 = **D**, 0011 = **3** • Answer: $(D3)_{16}$	• Convert $(11011.01)_2$ to Hex. • Whole part (group from right to left): **1** | **1011** →pad →**0001** | **1011** • Fraction part (group from left to right): **01** →pad →**0100** • Convert: 0001 = **1**, 1011 = **B** . 0100 = **4** • Answer: $(1B.4)_{16}$

CONVERSION FROM BINARY TO HEX

- **Practice Problem:** Convert these binary numbers to Hex. *(Hint: Group by 4 and watch out for padding!)*

Whole Numbers	Fractional Numbers
$(1111)_2$ $(11001)_2$ $(10101010)_2$ $(11100011)_2$ $(100000000)_2$	$(101.11)_2$ $(1110.1)_2$